

STUDENT UTILITY PROGRAM

Experiment No: _____ Date: _____

Title: Implementation of Student Utility Operations using Java Classes and Objects

OBJECTIVE / AIM

To write a Java program using classes and objects to perform various student utility operations such as student information, even or odd checking, finding the largest number, grade calculation, day of the week, multiplication table, natural numbers, Fibonacci series, factorial, prime number checking, maximum of two numbers, and area of a circle.

SYSTEM REQUIREMENTS

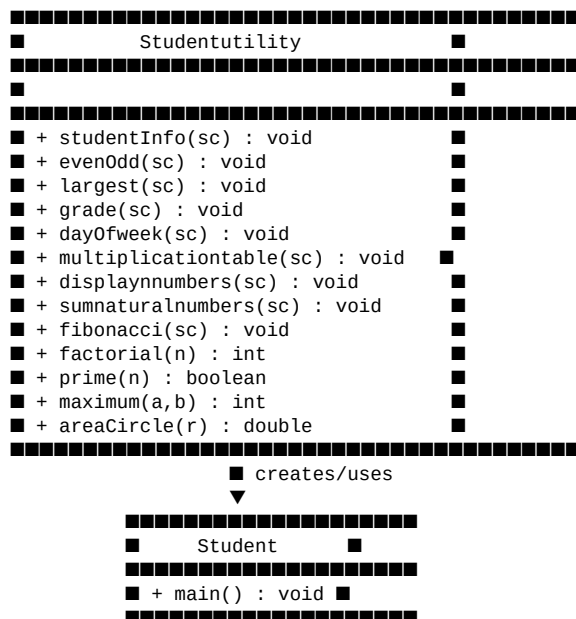
Hardware: Standard PC setup (Core i3/i5 processor, 4GB/8GB RAM).

Software: Windows/Linux/macOS, JDK 17 or JDK 21, and an IDE or text editor such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code.

ALGORITHM / LOGIC DESCRIPTION

1. Start the program.
2. Import the Scanner class for user input.
3. Create a Studentutility class containing methods for all utility operations.
4. Create an object of Studentutility in the main method.
5. Display a menu containing 14 choices.
6. Read the user's choice using Scanner.
7. Execute the corresponding method using a switch statement.
8. Repeat the menu until the user selects Exit.
9. Display the required result for each operation.
10. Stop the program.

UML CLASS DIAGRAM



SOURCE CODE

```
package mypackage;
import java.util.Scanner;
class Studentutility {
```

```

void studentInfo(Scanner sc) {
    System.out.print("Enter Student Name : ");
    String name = sc.nextLine();

    System.out.print("Enter Roll Number : ");
    int roll = sc.nextInt();

    System.out.print("Enter Marks 1 : ");
    int m1 = sc.nextInt();

    System.out.print("Enter Marks 2 : ");
    int m2 = sc.nextInt();

    System.out.print("Enter Marks 3 : ");
    int m3 = sc.nextInt();

    int total = m1 + m2 + m3;
    double percentage = total / 3.0;

    System.out.println("\nStudent Details");
    System.out.println("Name : " + name);
    System.out.println("Roll No : " + roll);
    System.out.println("Total : " + total);
    System.out.println("Percentage : " + percentage);
}

void evenOdd(Scanner sc) {
    System.out.print("Enter a Number : ");
    int n = sc.nextInt();

    if (n % 2 == 0)
        System.out.println("Even Number");
    else
        System.out.println("Odd Number");
}

void largest(Scanner sc) {
    System.out.print("Enter First Number : ");
    int a = sc.nextInt();

    System.out.print("Enter Second Number : ");
    int b = sc.nextInt();

    System.out.print("Enter Third Number : ");
    int c = sc.nextInt();

    if (a >= b && a >= c)
        System.out.println("Largest = " + a);
    else if (b >= a && b >= c)
        System.out.println("Largest = " + b);
    else
        System.out.println("Largest = " + c);
}

void grade(Scanner sc) {
    System.out.print("Enter Percentage : ");
    double p = sc.nextDouble();

    if (p >= 90)
        System.out.println("Grade A");
    else if (p >= 75)
        System.out.println("Grade B");
    else if (p >= 60)
        System.out.println("Grade C");
    else if (p >= 40)
        System.out.println("Grade D");
    else
        System.out.println("Fail");
}

void dayOfWeek(Scanner sc) {
    System.out.print("Enter Day Number (1-7) : ");
    int day = sc.nextInt();

    switch (day) {
        case 1: System.out.println("Monday"); break;
        case 2: System.out.println("Tuesday"); break;
        case 3: System.out.println("Wednesday"); break;
        case 4: System.out.println("Thursday"); break;
        case 5: System.out.println("Friday"); break;
        case 6: System.out.println("Saturday"); break;
        case 7: System.out.println("Sunday"); break;
        default: System.out.println("Invalid Day");
    }
}

```

```

    }
}

void multiplicationtable(Scanner sc) {
    System.out.print("Enter Number : ");
    int n = sc.nextInt();

    for (int i = 1; i <= 10; i++)
        System.out.println(n + " x " + i + " = " + (n * i));
}

void displaynnumbers(Scanner sc) {
    System.out.print("Enter N : ");
    int n = sc.nextInt();

    for (int i = 1; i <= n; i++)
        System.out.print(i + " ");

    System.out.println();
}

void sumnaturalnumbers(Scanner sc) {
    System.out.print("Enter N : ");
    int n = sc.nextInt();

    int sum = 0;
    for (int i = 1; i <= n; i++)
        sum += i;

    System.out.println("Sum = " + sum);
}

void fibonacci(Scanner sc) {
    System.out.print("Enter Number of Terms : ");
    int n = sc.nextInt();

    int a = 0, b = 1;
    System.out.print("Fibonacci Series : ");

    for (int i = 1; i <= n; i++) {
        System.out.print(a + " ");
        int c = a + b;
        a = b;
        b = c;
    }

    System.out.println();
}

int factorial(int n) {
    int fact = 1;

    for (int i = 1; i <= n; i++)
        fact *= i;

    return fact;
}

boolean prime(int n) {
    if (n <= 1)
        return false;

    for (int i = 2; i <= n / 2; i++) {
        if (n % i == 0)
            return false;
    }

    return true;
}

int maximum(int a, int b) {
    if (a > b)
        return a;
    else
        return b;
}

double areaCircle(double r) {
    return 3.14 * r * r;
}
}

public class Student {

```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    Studentutility obj = new Studentutility();
    int choice;

    do {
        System.out.println("\n===== STUDENT UTILITY PROGRAM =====");
        System.out.println("1. Student Information");
        System.out.println("2. Even or Odd");
        System.out.println("3. Largest of Three Numbers");
        System.out.println("4. Display Grade");
        System.out.println("5. Day of the Week");
        System.out.println("6. Multiplication Table");
        System.out.println("7. Display Numbers");
        System.out.println("8. Sum of Natural Numbers");
        System.out.println("9. Fibonacci Series");
        System.out.println("10. Factorial");
        System.out.println("11. Prime Number");
        System.out.println("12. Maximum of Two Numbers");
        System.out.println("13. Area of Circle");
        System.out.println("14. Exit");

        System.out.print("Enter Your Choice : ");
        choice = sc.nextInt();
        sc.nextLine();

        switch (choice) {
            case 1: obj.studentInfo(sc); break;
            case 2: obj.evenOdd(sc); break;
            case 3: obj.largest(sc); break;
            case 4: obj.grade(sc); break;
            case 5: obj.dayOfWeek(sc); break;
            case 6: obj.multiplicationtable(sc); break;
            case 7: obj.displaynnumbers(sc); break;
            case 8: obj.sumnaturalnumbers(sc); break;
            case 9: obj.fibonacci(sc); break;

            case 10:
                System.out.print("Enter Number : ");
                int n = sc.nextInt();
                System.out.println("Factorial = " + obj.factorial(n));
                break;

            case 11:
                System.out.print("Enter Number : ");
                n = sc.nextInt();

                if (obj.prime(n))
                    System.out.println("Prime Number");
                else
                    System.out.println("Not a Prime Number");
                break;

            case 12:
                System.out.print("Enter First Number : ");
                int a = sc.nextInt();

                System.out.print("Enter Second Number : ");
                int b = sc.nextInt();

                System.out.println("Maximum = " + obj.maximum(a, b));
                break;

            case 13:
                System.out.print("Enter Radius : ");
                double r = sc.nextDouble();

                System.out.println("Area = " + obj.areaCircle(r));
                break;

            case 14:
                System.out.println("Thank You...");
                break;

            default:
                System.out.println("Invalid Choice");
        }
    } while (choice != 14);
}

```

COMPILATION & EXECUTION COMMANDS

Save the file as Student.java

Compile:
javac -d . Student.java

Run:
java mypackage.Student

OUTPUT SCREEN

Sample console output:

```
===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Display Grade
5. Day of the Week
6. Multiplication Table
7. Display Numbers
8. Sum of Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter Your Choice : 2
Enter a Number : 10
Even Number

===== STUDENT UTILITY PROGRAM =====
Enter Your Choice : 10
Enter Number : 5
Factorial = 120

===== STUDENT UTILITY PROGRAM =====
Enter Your Choice : 14
Thank You...
```